

Day 28 — Decision Tree Classifier

Veda Technology · Level 2 · Day 28

Objective: Train a Decision Tree classifier and explore how tree depth and other parameters influence model performance.

Dataset: Iris Dataset from scikit-learn.

1. Workflow

- Load and inspect Iris data
- Train-test split
- Train Decision Tree
- Visualize the tree
- Compare max_depth values 1–10
- Analyze training/testing accuracy
- Inspect feature importance
- Study overfitting

2. Core Code

```
model = DecisionTreeClassifier(max_depth=3, random_state=42)
model.fit(X_train, y_train)

train_pred = model.predict(X_train)
test_pred = model.predict(X_test)

train_accuracy = accuracy_score(y_train, train_pred)
test_accuracy = accuracy_score(y_test, test_pred)
```

3. Tree Depth

The notebook tests depths from 1 to 10. Training and testing accuracy are compared to study underfitting and overfitting.

4. Feature Importance

The model's feature_importances_ values show which features contributed most to the tree's splits.

5. Interview Questions

How does a decision tree make predictions? It follows feature-based split rules from the root to a leaf.

What is overfitting? A tree learns training-specific patterns and performs worse on unseen data.

What does max_depth control? The maximum tree depth and model complexity.

6. Conclusion

This project demonstrates Decision Tree classification, visualization, depth tuning, feature importance, and overfitting analysis.